# Scalability & Future Plan

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| **Date** | 5 July 2026 |
| **Team ID** | XXXXXX |
| **Project Name** | OptiCrop: Smart Agricultural Production Optimization Engine |
| **Maximum Marks** | 1 Mark |

## Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

## Current System Limitations:

| S.No | Limitation | Impact | Priority to Address (High / Medium / Low) |
|---|---|---|---|
| 1 | Runs on local server only | Cannot support large number of users | High |
| 2 | Limited dataset for prediction model | Affects accuracy in real-world scenarios | High |
| 3 | No cloud deploymen | Accessibility is restricted to local system | Medium |

## Scalability Plan:

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User Load | Supports limited users (local testing) | Deploy on cloud (AWS / Azure / Render) |
| 2 | Data Storage | Local database (SQLite/MySQL) | Upgrade to scalable cloud database (MongoDB Atlas / Firebase) |
| 3 | Performance | Moderate response time under load | Implement caching and load balancing |
| 4 | Security | Basic authentication | Add JWT authentication and encryption |

## Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|

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|---|---|---|---|
| Phase 2 | Mobile application version | 2–3 months | Easier farmer access via smartphones |
| Phase 3 | AI-based crop disease detection | 4–6 months | Improve crop health monitoring |
| Phase 4 | Market price prediction system | 6–8 months | Help farmers maximize profit |